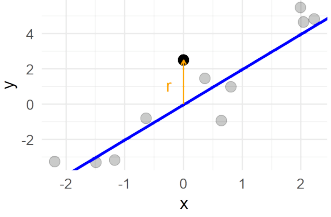
Introduction to Machine Learning

Advanced Risk Minimization
Pseudo-Residuals



Learning goals

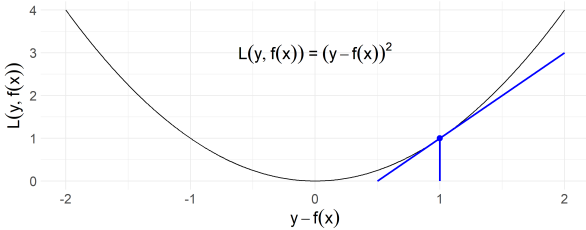
- Concept of pseudo-residuals
- PRs for common losses

PSEUDO-RESIDUALS

- In regression, residuals are defined as $r(\mathbf{x}) := y - f(\mathbf{x})$
- Generalize concept to **pseudo-residuals**:

$$\tilde{r}(\mathbf{x}) := -\frac{\partial L\left(y, f(\mathbf{x})\right)}{\partial f(\mathbf{x})}$$

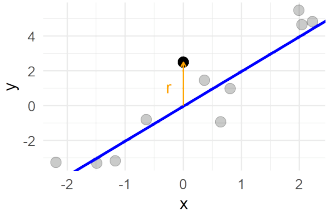
- If we wiggle $f(\mathbf{x})$, how much does L change?
- Can be used for score-based classifiers and other models
- Note that $\tilde{r}(\mathbf{x})$ depends on y, $f(\mathbf{x})$ and L



BEST POINT-WISE UPDATE

- Assume we have (partially) fitted a model $f(\mathbf{x})$ to data $\mathcal{D}$
- Assume we could update $f(\mathbf{x})$ point-wise as we like
- Under squared loss, for a fixed $\mathbf{x} \in \mathcal{X}$, the best point-wise update is the direction of the residual $r(\mathbf{x}) = y - f(\mathbf{x})$

$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) + r(\mathbf{x})$$



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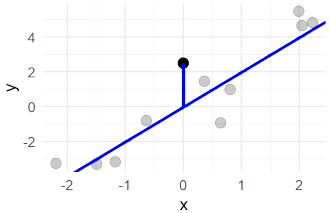
- Point-wise error at this specific $\mathbf{x}$ becomes 0



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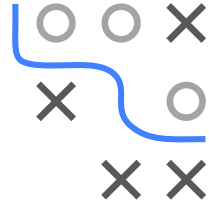

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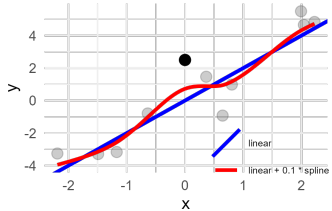
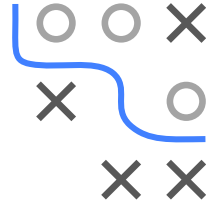
- In gradient boosting, we don't do point-wise updates but "smoothly distort" f so we generalize



APPROXIMATE BEST POINT-WISE UPDATE



- When applying GD to compute a point-wise update of $f(\mathbf{x})$, we go a step of negative gradient of loss

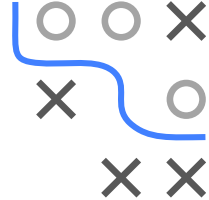
$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) - \frac{\partial L\left(y, f(\mathbf{x})\right)}{\partial f(\mathbf{x})}$$

- This would be for one observation
- This is effectively the PR which is the direction of the pseudo-residual

$$f(\mathbf{x}) \quad \leftarrow \quad f(\mathbf{x}) + \tilde{r}(\mathbf{x})$$

- Iteratively stepping towards the direction of the pseudo-residuals is the underlying idea of gradient boosting, which is a learning algorithm that will be covered in a later chapter

GD IN ML AND PSEUDO-RESIDUALS



- In GD, we move in the direction of the negative gradient by updating the parameters:

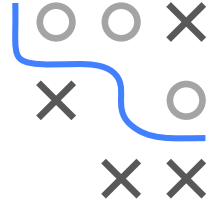
$$\boldsymbol{\theta}^{[t+1]} = \boldsymbol{\theta}^{[t]} - \alpha^{[t]} \cdot \nabla_{\boldsymbol{\theta}} \, \mathcal{R}_{\text{emp}}(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\boldsymbol{\theta}^{[t]}}$$

- This can be seen as approximating the unexplained information (measured by the loss) through a model update
- Using the chain rule:

$$\begin{aligned} \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) &= \sum_{i=1}^{n} \frac{\partial L\left(y^{(i)}, f(\mathbf{x})\right)}{\partial f(\mathbf{x})}\bigg|_{f=f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)} \cdot \nabla_{\boldsymbol{\theta}} f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right) \\ &= -\sum_{i=1}^{n} \tilde{r}^{(i)} \cdot \nabla_{\boldsymbol{\theta}} f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right). \end{aligned}$$

- Update is loss-optimal directional change of model output and a loss-independent derivative of $f(\mathbf{x})$
- This is a flexible, nearly loss-independent variant of GD

PSEUDO-RESIDUALS FOR COMMON LOSSES



Loss	**Domain of y**	**Pseudo residual $\tilde{r}$**
Squared loss	$y \in \mathbb{R}$	$y - f(\mathbf{x})$
Bernoulli loss	$y \in \{0, 1\}$	$y - s(f(\mathbf{x})) = y - \pi(\mathbf{x})$
Multinomial loss	$y \in \{1, \dots, g\}$	$\mathbb{1}_{\{y=k\}} - \pi_k(\mathbf{x})$
Exponential loss	$y \in \{-1, 1\}$	$y \exp(-yf(\mathbf{x}))$

NB: $\pi(\mathbf{x}) = s(f(\mathbf{x})) = \frac{\exp(f(\mathbf{x}))}{1+\exp(f(\mathbf{x}))}$ is the (sigmoidal) logistic function, and $\pi_k(\mathbf{x})$ its multi-class generalization, the softmax.